

We can use **Cloudflare Tunnel** to expose a web service running on a local port, either on Rivanna or on your personal device, through a temporary public URL so others can access it in a browser.

Since compute nodes on UVA Rivanna/your own pc are not directly exposed to the public internet, we do the following:

1. Run your agent server on Rivanna/your own pc, listening on port **xxxx**
2. Start a **Cloudflare tunnel** with **cloudflare**
3. Cloudflare generates a temporary public URL(for testing only) like:

<https://xxxx.trycloudflare.com>

So the connect flow is :

User Browser → Public Cloudflare URL → Cloudflare Tunnel → Rivanna/personal compute node(<http://127.0.0.1:xxxx>)

To make it public, quick

- 1: Run your service on Rivanna/your own pc

For example host on the port 3000 : <http://localhost:3000>

- 2: Install Cloudflare Tunnel

```
wget https://github.com/cloudflare/cloudflared/releases/latest/download/cloudflared-linux-amd64
chmod +x cloudflared-linux-amd64
mv cloudflared-linux-amd64 cloudflared
```

Make sure the installation is successful:

`./cloudflared --version`

You should see the version number printed.

- 3.Login to Cloudflare

`./cloudflared tunnel login`

4. Using tunnel to public local host(in this example the port is 3000)

`./cloudflared tunnel --url http://localhost:3000`

Click for more details via <https://developers.cloudflare.com/tunnel/setup/>